

HOSPITAL MANAGEMENT SYSTEM

Online Appointment Booking

A PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF
THE DEGREE OF

B.TECH. (COMPUTER ENGINEERING)
TO
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SUBMITTED BY

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August 2026



SCHOOL OF ENGINEERING, RK UNIVERSITY, RAJKOT

CERTIFICATE

This is to certify that the work which is being presented in the Project Report entitled "**KK PATEL HOSPITAL Website in Laravel**", in partial fulfillment of the requirement for the award of the degree of B.Tech. (Computer Engineering) and submitted to the School of Engineering, RK University, is an authentic record of my/our own work carried out during a period from june 2026 to November 2026.

The matter presented in this Project Report has not been submitted by me/us for the award of any other degree elsewhere.

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DECLARATION

I/We hereby certify that I/We am/are the sole author(s) of this project work and that neither any part of this project work nor the whole of the project work has been submitted for a degree to any other University or Institution.

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ABSTRACT

KK Patel Hospital Website is an online platform that allows guest users to book appointments directly from their laptops or mobile phones. This website typically includes features such as Available Departments, About the Hospital, Online Appointment Booking, etc. It is a fully responsive website. A confirmation email is sent upon that your appointment is successful or rejected.

Keywords:Access all the facilities, Departments, Online appointment booking, Free to Contact us, give feedbacks online etc.

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Chapter 1
INTRODUCTION

1.1 Project Summary

KK Patel Super Speciality Hospital is a premier multi-speciality tertiary healthcare institution located in Bhuj, Kachchh, Gujarat. Managed with exemplary stewardship by Shree Kutch Leva Patel Kanya Kelavani Mandal, the hospital is built upon a philosophy of providing state-of-the-art medical care with human compassion. Established to address the growing regional demand for advanced diagnostics, cardiology, oncology, nephrology, and general medicine, the medical center combines modern medical technologies with a dedicated team of accredited medical specialists.

The **KK Patel Hospital Web Application and Clinical Management System** is a comprehensive, enterprise-grade digital health solution developed using the modern Laravel PHP framework. The system eliminates historical bottlenecks associated with manual outpatient department (OPD) queueing, paper token issuance, uncoordinated physician schedules, and paper prescription tracking. It seamlessly unifies four discrete user environments: a public-facing informative portal, an authenticated patient portal, a dedicated clinical operations console for specialist doctors, and an executive administration suite.



Figure 1.1: KK Patel Super Speciality Hospital Infrastructure

1.2 Aim and Objectives

The primary aim of the KK Patel Hospital digital platform is to transform hospital operations from traditional, paper-intensive queues into an automated, highly secure, and auditable digital health ecosystem. The targeted objectives include:

- **Streamlined Online OPD Scheduling:** Provide intuitive, 24/7 web-based appointment reservation capabilities allowing patients to select clinical departments, review specialist qualifications, and book slots in real-time.
- **Real-Time Slot Quota & Off-Duty Automation:** Enforce strict per-slot patient capacities, eliminate outpatient overbooking, and dynamically reflect doctor availability status based on weekly shifts and scheduled off-days.
- **Digital OPD Token Slips:** Generate tamper-evident, standardized consultation vouchers complete with Unique Patient Identifiers (#PAT-XXXXX), consultation tokens (#KKP-XXXXX), QR verification barcodes, and 1-click PDF download.
- **Clinical Prescription Authoring:** Empower attending doctors to create structured digital medical prescriptions incorporating chief complaints, primary diagnoses, multi-drug pharmaceutical regimens, dosage frequencies, and follow-up guidance.
- **Data Security & Legal Integrity:** Implement rigorous role-based access control (RBAC), multi-guard authentication, medical integrity locks (preventing unauthorized alteration of medical prescriptions), and patient privacy protections.

1.3 Problem Statement & Healthcare Challenges

Historically, district hospitals in western Gujarat have faced significant operational challenges due to manual patient processing. Patients often travel long distances only to discover that their designated specialist is off-duty, or they encounter overwhelming OPD queues leading to hours of waiting. Furthermore, handwritten paper consultation slips and prescriptions present readability hazards, risk of pharmaceutical dispensing errors, and are easily lost by patients. Hospital management previously lacked centralized real-time visibility into consultation throughput, doctor workload distribution, and department patient flow.

1.4 Scope of the Project

The project encompasses the complete operational spectrum of outpatient healthcare management across four primary functional portals:

- **1. Public Healthcare Portal:** Informational hub delivering hospital background, accredited doctor directory, clinical wing capabilities, press releases, emergency helpline numbers, and guest appointment booking.
- **2. Patient / User Portal:** Secured patient self-service account with automated profile management, live consultation booking, booking history tracking, downloadable digital OPD slips, and view/download access to issued digital prescriptions.
- **3. Doctor Clinical Operations Console:** High-security clinical interface enabling specialist doctors to monitor their daily consultation queues, update appointment outcomes (Approved, Completed, Pending, Cancelled), manage daily time slot quotas, configure weekly shift availability, and author full digital prescriptions.
- **4. Hospital Administration Suite:** Central governance console managing doctor roster allocations, departmental configurations, master appointment schedules, patient reviews, press desk releases, and medical integrity audit logs.

1.5 Stakeholders & Target Audience

The system is engineered for diverse user archetypes across the regional healthcare network:

- **Outpatients & Families:** Seeking swift, dependable access to specialist care without unnecessary hospital physical queueing.
- **Attending Medical Specialists:** Requiring streamlined queue management, accurate time slot quotas, and rapid prescription generation tools.
- **Hospital Medical Registrars & Desk Staff:** Responsible for validating patient arrival, verifying OPD voucher tokens, and directing patients to allotted consultation rooms.
- **Executive Hospital Management & Trustees:** Overseeing institutional operational performance, department utilization, and clinical service excellence.

Chapter 2
SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The application is designed to operate efficiently across modern commodity server hardware and diverse end-user client devices:

Hardware Tier	Minimum Specifications	Recommended Production Specifications
Web / Application Server	Intel Core i3 / Xeon 2.4 GHz, 4 GB RAM, 60 GB SSD	Intel Xeon 8-Core / AMD EPYC 3.0 GHz, 16 GB DDR4 RAM, NVMe SSD Storage
Database Server	Dual-Core 2.0 GHz, 4 GB RAM, 40 GB Storage	Quad-Core Xeon 3.2 GHz, 16 GB RAM, High-IOPS RAID NVMe Storage
Client Desktop / Laptop	Core 2 Duo / 2 GB RAM, 1024x768 Display	Intel Core i5 / 8 GB RAM, Full HD 1920x1080 Display
Mobile / Tablet Devices	Android 8.0+ or iOS 12.0+, 2 GB RAM, 4G/WiFi	Android 12+ or iOS 16+, 4 GB RAM, High-Speed Broadband Connectivity

2.2 Server Requirements & Technology Stack

- **Operating System:** Windows Server / Ubuntu Linux 22.04 LTS.
- **Web Server Engine:** Apache HTTP Server 2.4+ / Nginx with URL rewriting enabled.
- **PHP Runtime:** PHP 8.2 or 8.3+ with extensions: OpenSSL, PDO, Mbstring, Tokenizer, XML, Ctype, JSON, BCMath, and Fileinfo.
- **Database Management System:** MySQL 8.0+ with InnoDB storage engine ensuring ACID transaction compliance.
- **Dependency Manager:** Composer 2.6+ for backend PHP packages.
- **Node.js & NPM:** Node.js v18+ and NPM 9+ for front-end asset compilation and bundling via Vite / Laravel Mix.

2.3 Web Browser Compatibility Matrix

To guarantee universal accessibility for patients booking appointments and medical doctors operating clinical consoles, the application adheres to W3C web standards and ECMAScript 6+ specifications. The compatibility matrix across modern browsers is certified below:

Web Browser	Engine	Minimum Version	Compatibility Status
Google Chrome	Blink	Version 90+	Fully Supported (Verified)
Microsoft Edge	Chromium	Version 90+	Fully Supported (Verified)
Mozilla Firefox	Gecko	Version 88+	Fully Supported (Verified)
Apple Safari	WebKit	Version 14+	Fully Supported (Verified)
Mobile Browsers	Chrome Mobile / Safari Mobile	iOS 13+ / Android 9+	Fully Supported (Responsive Layout)

2.4 Software Packages & Development Stack

• **Laravel Framework (10.x / 11.x):** Robust MVC architecture, Eloquent ORM, Blade templating engine, routing, and database migrations.

• **Bootstrap 4 / 5 & Custom CSS:** Responsive grid framework styled with modern medical color palettes (KK Patel Purple #69275c, Clinical Teal #00a896, and Medical Blue #0284c7).

• **jQuery & AJAX Engine:** High-performance asynchronous DOM manipulation enabling dynamic doctor dropdown fetching, real-time slot checking, and token validation.

• **html2pdf.js:** Client-side canvas rendering library generating high-resolution, vector-accurate PDF documents for OPD consultation slips and medical prescriptions.

• **Visual Studio Code & Git:** Industry-standard code editor with integrated Git version control for repository management.

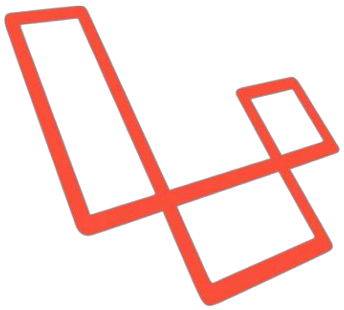


Figure 2.1: Laravel MVC Framework Technology Architecture

Chapter 3
SYSTEM ANALYSIS

3.1 Feasibility Study

A rigorous feasibility study was performed to assess the operational, economic, technical, and scheduling viability of developing an end-to-end web management system for KK Patel Super Speciality Hospital.

1. Technical Feasibility: The required technological ecosystem (Laravel, PHP, MySQL, Apache, Bootstrap, and JavaScript) consists of mature, battle-tested open-source technologies with vast community support. The development team possessed verified expertise in Laravel MVC, Eloquent ORM, and database schema optimization. Server hosting costs and hardware specifications are well within standard web hosting tiers, confirming high technical feasibility.

2. Economic Feasibility: Transitioning from manual paper-based token slips and physical registers to an automated web solution significantly diminishes recurrent stationery costs, administrative overheads, and manual scheduling errors. The economic return on investment (ROI) is achieved through improved clinical throughput, patient retention, zero software licensing expenses (open-source stack), and elimination of duplicate booking chaos.

3. Operational Feasibility: Hospital staff, reception registrars, and clinical specialists require minimal operational training due to clean, intuitive user interfaces tailored to each role. Real-time availability toggles and 1-click status changers integrate seamlessly into existing clinical workflows.

4. Schedule Feasibility: The project development schedule spanned June 2026 to November 2026, comprising systematic sprint milestones: requirement elicitation, database architecture, multi-guard authentication, slot quota automation, prescription engine implementation, and comprehensive testing.

3.2 Technical Analysis & MVC Architecture in Laravel

The application leverages the Model-View-Controller (MVC) architectural pattern provided by Laravel. MVC establishes a strict separation of concerns:

• **Model Layer (M):** Encapsulates business logic, data models, and database interactions using Laravel's Eloquent ORM. Models include *User*, *Doctors*, *Department*, *appointment*, *DoctorTimeSlot*, *DoctorSchedule*, *Prescription*, and *PrescriptionMedicine*. Active relationships (hasOne, hasMany, belongsTo) ensure relational data integrity.

• **View Layer (V):** Constructed with Laravel Blade templating engine. The views employ layout inheritance (@extends, @section, @stack) ensuring consistent navigation, responsive CSS grids, and real-time DOM elements across doctor, patient, and admin panels.

• **Controller Layer (C):** Serves as the cognitive bridge, processing HTTP requests, executing database transactions, validating form inputs, and returning formatted view responses or JSON payloads for asynchronous AJAX calls.

3.3 Functional Requirements Analysis

Functional requirements are categorized across the four user subsystems:

Module 1: Public Portal: Browse hospital facilities, explore medical departments, search doctors by speciality, submit inquiries via Contact form, and book outpatient consultations as a guest.

Module 2: Patient Portal: Patient registration and login, manage demographic profile, view live remaining slots per doctor, book appointments, view digital OPD slips with barcodes, and download finalized medical prescriptions as PDF.

Module 3: Doctor Clinical Operations: Secure doctor authentication guard, review daily consultation rosters, update consultation statuses, configure weekly time slots and maximum patient quotas, toggle on-duty/off-duty availability, and author full digital prescriptions.

Module 4: Prescription Engine: Capture chief symptoms, symptom duration, primary clinical diagnosis, clinical vitals, medical history, known drug allergies, dynamic medication tables (name, dosage, frequency, duration, route, instructions), advice, and follow-up consultation dates.

Module 5: Admin Command Suite: Full CRUD over doctors and medical departments, master view of all booked appointments, quick status updates, prescription audit logs with medical integrity lock.

3.4 Non-Functional Requirements

Non-functional requirements govern the behavioral quality, resilience, and operational parameters of the healthcare system:

- **1. Data Security & Role Isolation:** Multi-guard authentication segregates patient, doctor, and administrator sessions. Passwords are encrypted using the Bcrypt hashing algorithm with high work factors. Sensitive medical records (prescriptions) are restricted strictly to the authoring doctor, the designated patient, and read-only administrative auditing.
- **2. Medical Integrity & Auditability:** Issued medical prescriptions are locked from unauthorized modification by administrative staff or doctors once finalized. All status transitions record timestamps in the database.
- **3. High Performance & Low Latency:** Server response times for doctor slot queries and patient appointment lookups are maintained below 150 milliseconds through database indexing and Eloquent eager-loading (preventing N+1 query overheads).
- **4. Responsive User Experience:** User interfaces dynamically adapt to all screen form factors (smartphones, tablets, and high-resolution clinical monitors) using responsive Bootstrap grids and mobile-optimized navigation toggles.
- **5. Reliability & Transaction Integrity:** Critical data write operations (such as saving a prescription and its associated medication rows) are wrapped within atomic database transactions (DB::transaction), guaranteeing that partial failures never leave orphaned records.

Chapter 4

SYSTEM DESIGN

4.1 System Architectural Design

The architecture follows a multi-tier, multi-guard paradigm. Incoming client requests pass through the HTTP web server, where Laravel's routing middleware inspects authentication tokens and applies role-based middleware guards (*web* for registered patients, *doctor.auth* for clinical doctors, and *checkAdmin* for hospital administrators). Controllers orchestrate the business logic, interacting with MySQL through Eloquent models and returning responsive Blade views.

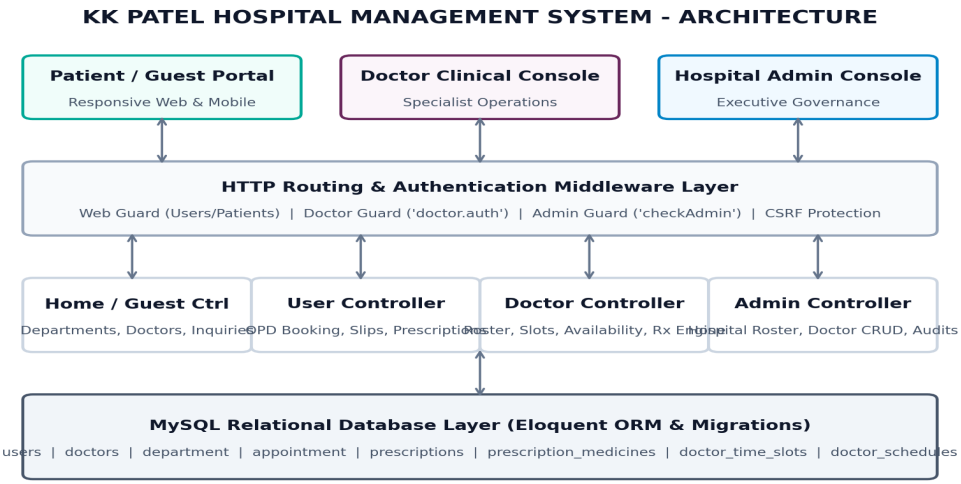


Figure 4.1: Layered MVC System Architecture & Multi-Guard Security

4.2 Data Flow Diagrams (DFD Level 0 & Level 1)

Data Flow Diagrams illustrate the trajectory of healthcare data as it moves between external entities (Patients, Doctors, Administrators), automated processes, and database repositories. In Level 0 (Context Level), external users submit credentials and consultation requests, receiving authenticated dashboards, OPD vouchers, and medical prescriptions. Level 1 delineates internal sub-processes:

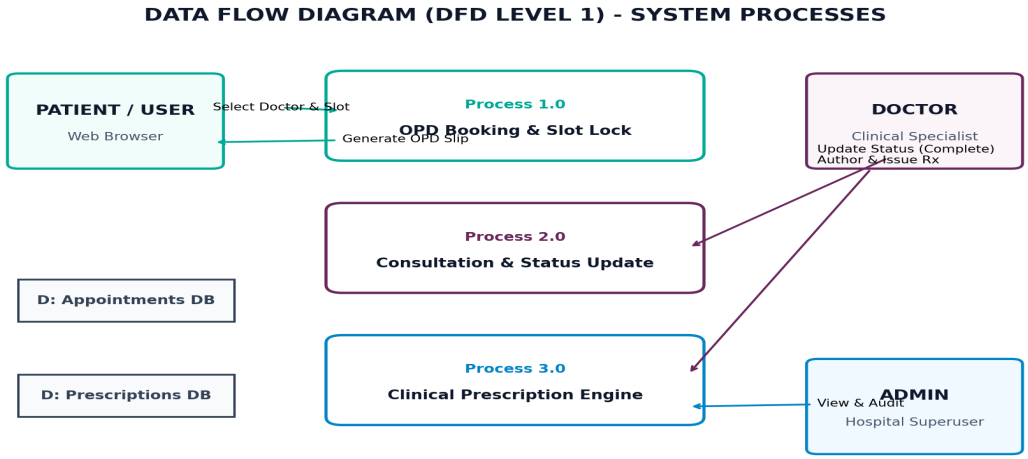


Figure 4.2: Data Flow Diagram (DFD Level 1) - Operational Processes

4.3 UML Use Case Modeling

The system defines four distinct actors interacting with specialized use cases:

• **Guest User:** View hospital overview, browse medical departments, view doctor bios, submit general inquiry messages, and book appointments via guest checkout.

• **Registered Patient:** Authenticate into patient portal, book real-time OPD appointments with doctor availability checks, track booking status, view/print digital OPD token slips, and download finalized clinical prescriptions.

• **Consulting Doctor:** Secure doctor login, view assigned consultation queue, filter consultations by date/status, update consultation outcomes (Approved, Completed, Cancelled), configure time slots with patient quotas, toggle on-duty/off-duty availability, author structured prescriptions with medicines, and issue finalized prescriptions.

• **Hospital Administrator:** Manage doctor directory, configure medical departments, oversee global appointment queue, inspect patient feedback, post press releases, and audit prescriptions under read-only integrity controls.

4.4 UML Activity Diagram & Clinical Consultation Lifecycle

The activity diagram models the end-to-end operational workflow spanning the patient booking phase, system quota verification, clinical consultation review, and prescription issuance:

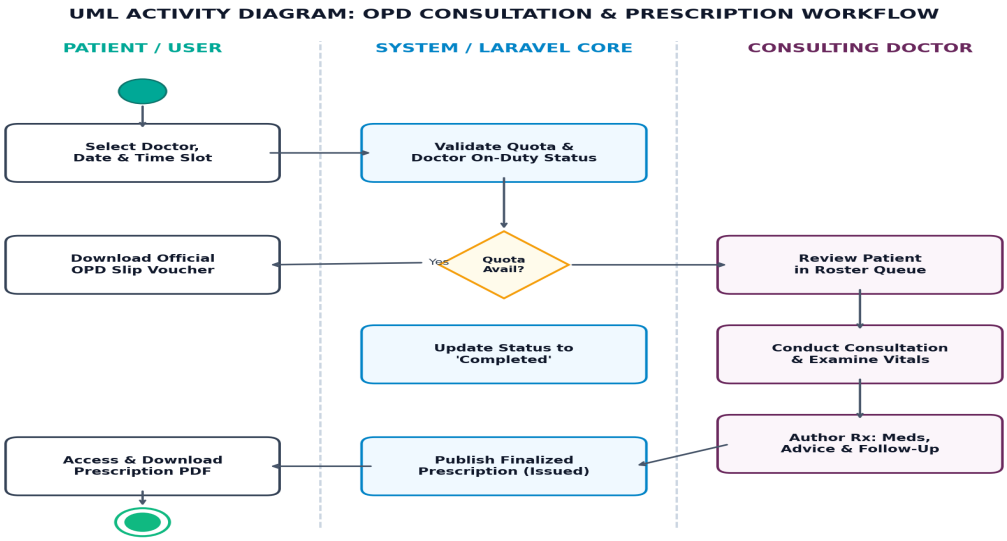


Figure 4.3: UML Activity Diagram - OPD Consultation & Prescription Lifecycle

4.5 Sequence Flow Diagrams

The sequence flow diagram illustrates chronological message exchanges between system objects during consultation authoring:

1. **Patient Initiates Booking:** Client browser submits appointment request with selected doctor and date. *UserController* invokes *DoctorTimeSlot* to compute remaining token capacity. If slot capacity is available and doctor is on-duty, appointment record is created in *appointment* table with status 'Approved'.

2. **Doctor Conducts Consultation:** Doctor accesses *DoctorController::appointments()*, reviews patient chief complaints, and marks consultation as 'Completed'.

3. **Prescription Authoring:** Completed consultation unlocks 'Add Prescription' action. Doctor enters symptoms, diagnosis, medicines, and advice. Submitting 'issue' executes a database transaction creating *Prescription* and child *PrescriptionMedicine* records.

4. **Patient Access:** Patient refreshes their portal, sees 'View Rx' action, and generates the official printable medical document via *html2pdf.js*.

Chapter 5
DATA DICTIONARY

5.1 Database Overview & Architecture

The database architecture is implemented in MySQL 8.0 with InnoDB relational storage engine. Foreign key constraints enforce referential integrity across users, appointments, doctor schedules, time slots, and prescriptions. All tables utilize indexed primary keys (auto-incrementing BIGINT) and UTF-8 Unicode collation (utf8mb4_unicode_ci) supporting international characters.

5.2 Entity Relationship Diagram (ERD)

The Entity-Relationship Diagram outlines the 12 interconnected tables in the hospital relational schema:

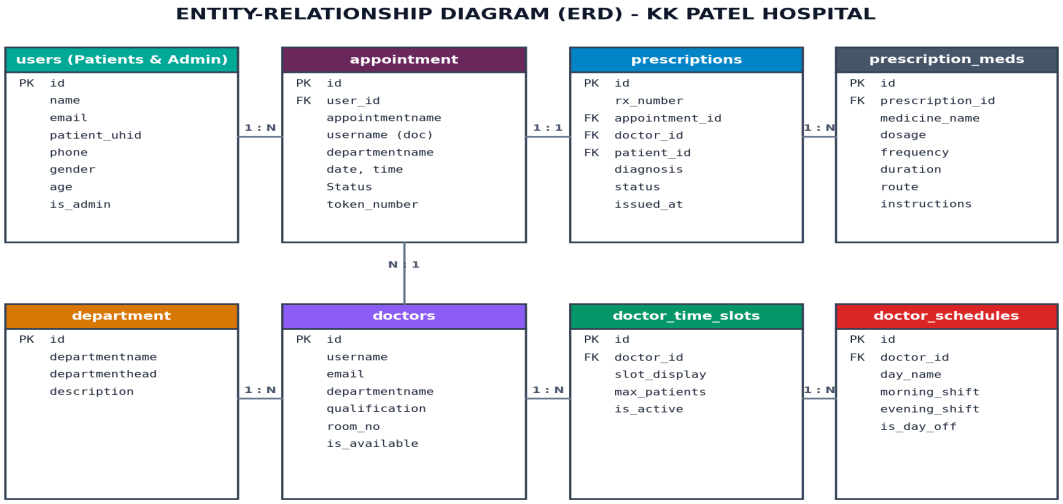


Figure 5.1: Entity-Relationship Diagram (ERD) - Relational Schema

5.3 Detailed Database Tables Schema & Explanations

Table 5.1: users Table (Patients & Administrators)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Unique user identification key
2	**name**	VARCHAR(255)	Required	Full legal name of patient or administrator
3	**email**	VARCHAR(255)	Unique, Required	Unique email address for portal login
4	**password**	VARCHAR(255)	Encrypted	Bcrypt hashed account authentication password
5	**patient_uhid**	VARCHAR(50)	Indexed, Nullable	Unique Hospital Identification Code (#PAT-XXXXX)
6	**phone**	VARCHAR(20)	Nullable	Patient emergency contact telephone number
7	**gender**	VARCHAR(20)	Nullable	Patient biological gender (Male, Female, Other)
8	**age**	INT	Nullable	Patient age in years
9	**is_admin**	TINYINT(1)	Default 0	Role flag: 1 for Hospital Admin, 0 for Patient
10	**created_at**	TIMESTAMP	Nullable	Account creation timestamp

Table 5.2: doctors Table (Specialist Medical Practitioners)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Doctor unique identification primary key
2	**username**	VARCHAR(255)	Unique, Required	Doctor full professional name (e.g. Dr. Meet Thacker)
3	**email**	VARCHAR(255)	Unique, Required	Doctor official clinical communication email
4	**password**	VARCHAR(255)	Encrypted	Hashed password for doctor console authentication
5	**departmentname**	VARCHAR(255)	Required	Assigned medical department wing
6	**qualification**	VARCHAR(255)	Nullable	Professional medical degrees (e.g. MBBS, MD)
7	**room_no**	VARCHAR(50)	Default 'OPD Room #102'	Designated physical outpatient consultation room
8	**consultation_fee**	DECIMAL(8,2)	Default 500.00	Hospital outpatient consultation fee
9	**is_available**	TINYINT(1)	Default 1	Master availability switch (1: On Duty, 0: Off Duty)

Table 5.3: department Table (Clinical Wings)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Department unique identification key
2	**departmentname**	VARCHAR(255)	Unique, Required	Name of clinical department (e.g. Cardiology)
3	**departmenthead**	VARCHAR(255)	Nullable	Head of Department (Senior Chief Consultant)
4	**departmentimage**	TEXT	Nullable	Image path for departmental banner visual
5	**Description**	TEXT	Nullable	Comprehensive clinical capabilities overview

Table 5.4: appointment Table (OPD Consultations)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Appointment consultation token ID (#KKP-XXXXX)
2	**user_id**	BIGINT UNSIGNED	FK -> users(id)	Foreign key linking registered patient account
3	**appointmentname**	VARCHAR(255)	Required	Patient full legal name on hospital record
4	**email**	VARCHAR(255)	Required	Patient email address for appointment alerts
5	**phone**	VARCHAR(20)	Required	Patient phone number for SMS and phone calls
6	**departmentname**	VARCHAR(255)	Required	Selected clinical department wing
7	**username**	VARCHAR(255)	Required	Assigned consulting specialist doctor name
8	**date**	DATE	Required	Allotted date for outpatient clinical consultation
9	**time**	VARCHAR(100)	Required	Assigned consultation time slot (e.g. 10:00 AM - 11:00 AM)
10	**Status**	VARCHAR(50)	Default 'Pending'	Consultation state: Approved, Completed, Cancelled
11	**message**	TEXT	Nullable	Chief presenting medical problem or reason for visit

Table 5.5: doctor_time_slots Table (Daily Capacity Quotas)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Time slot primary identification key
2	**doctor_id**	BIGINT UNSIGNED	FK -> doctors(id)	Foreign key linking associated specialist doctor
3	**slot_display**	VARCHAR(100)	Required	Formatted time label (e.g. 09:00 AM - 10:00 AM)
4	**max_patients**	INT	Default 8	Maximum patient quota limit per hourly time slot
5	**is_active**	TINYINT(1)	Default 1	Slot availability toggle (1: Active, 0: Inactive)
6	**day_of_week**	VARCHAR(20)	Default 'All Days'	Applicable recurrence day or daily assignment

Table 5.6: doctor_schedules Table (Weekly Shifts & Off-Days)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Doctor schedule primary key
2	**doctor_id**	BIGINT UNSIGNED	FK -> doctors(id)	Foreign key linking doctor profile
3	**day_name**	VARCHAR(20)	Required	Weekday name (Monday through Sunday)
4	**morning_start**	TIME	Default '09:00:00'	Morning OPD consultation shift start time
5	**morning_end**	TIME	Default '13:00:00'	Morning OPD consultation shift end time
6	**evening_start**	TIME	Default '16:00:00'	Evening OPD consultation shift start time
7	**evening_end**	TIME	Default '19:30:00'	Evening OPD consultation shift end time
8	**is_day_off**	TINYINT(1)	Default 0	Weekly scheduled rest day flag (1: Off Duty, 0: Working)

Table 5.7: prescriptions Table (Clinical Diagnoses & Advice)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Prescription primary identification key
2	**prescription_number**	VARCHAR(50)	Unique, Required	Official medical Rx voucher number (#RX-YYYY-XXXXX)
3	**appointment_id**	BIGINT UNSIGNED	FK -> appointment(id)	1-to-1 link to completed consultation appointment
4	**doctor_id**	BIGINT UNSIGNED	FK -> doctors(id)	Consulting specialist doctor authoring prescription
5	**patient_id**	BIGINT UNSIGNED	FK -> users(id)	Patient user account receiving prescription
6	**symptoms**	TEXT	Nullable	Documented patient presenting clinical complaints
7	**symptoms_duration**	VARCHAR(100)	Nullable	Duration of symptoms prior to consultation
8	**diagnosis**	TEXT	Required	Primary professional medical clinical diagnosis
9	**clinical_notes**	TEXT	Nullable	Physical examination vitals and clinical observations
10	**allergies**	TEXT	Nullable	Known patient drug or pharmaceutical allergies
11	**medical_advice**	TEXT	Nullable	Dietary, lifestyle, and rehabilitation clinical advice
12	**follow_up_required**	TINYINT(1)	Default 0	Flag indicating whether return visit is required
13	**follow_up_date**	DATE	Nullable	Allotted date for next follow-up evaluation
14	**status**	VARCHAR(20)	Default 'draft'	Prescription state: 'draft' or 'issued'
15	**issued_at**	TIMESTAMP	Nullable	Official finalized medical issuance timestamp

Table 5.8: prescription_medicines Table (Prescribed Pharmaceuticals)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Prescribed medicine primary key
2	**prescription_id**	BIGINT UNSIGNED	FK -> prescriptions(id)	Cascaded foreign key linking parent prescription
3	**medicine_name**	VARCHAR(255)	Required	Brand and pharmaceutical formulation name
4	**dosage**	VARCHAR(100)	Nullable	Prescribed dosage strength (e.g. 500mg / 1 Tablet)
5	**frequency**	VARCHAR(100)	Nullable	Administration schedule (e.g. 1-0-1 After Meals)
6	**duration**	VARCHAR(100)	Nullable	Treatment course duration (e.g. 5 Days, 1 Month)
7	**route**	VARCHAR(100)	Default 'Oral'	Administration route: Oral, Topical, IV, Inhalation
8	**instructions**	VARCHAR(255)	Nullable	Patient instructions (e.g. Take with warm water)

Table 5.9: feedback Table (Patient Reviews & Experience Ratings)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Feedback unique primary key
2	**name**	VARCHAR(255)	Required	Full name of patient submitting feedback review
3	**email**	VARCHAR(255)	Required	Contact email address of reviewer
4	**message**	TEXT	Required	Patient satisfaction feedback and experience notes
5	**created_at**	TIMESTAMP	Nullable	Submission timestamp

Table 5.10: contacts Table (Patient Inquiries & Grievances)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Contact inquiry primary key
2	**contact name**	VARCHAR(255)	Required	Inquirer full name
3	**email**	VARCHAR(255)	Required	Inquirer email address for response
4	**phone**	VARCHAR(20)	Required	Contact phone number
5	**subject**	VARCHAR(255)	Required	Subject heading of inquiry or grievance
6	**message**	TEXT	Required	Inquiry description or grievance details
7	**replied**	VARCHAR(20)	Default 'No'	Resolution status flag ('Yes' or 'No')

Table 5.11: press_desk Table (Hospital Media & News Releases)

#	Column Name	Data Type	Constraints	Description
1	**id**	BIGINT UNSIGNED	PK, Auto Increment	Media release primary key
2	**Press image**	VARCHAR(255)	Required	Server file storage path for news clipping image
3	**Description**	TEXT	Required	Press release editorial summary and media notes
4	**created_at**	TIMESTAMP	Nullable	Publication timestamp

Table 5.12: password_resets Table (Account Security Tokens)

#	Column Name	Data Type	Constraints	Description
1	**email**	VARCHAR(255)	Indexed, Required	User email address requesting password reset
2	**token**	VARCHAR(255)	Required	Cryptographically generated secure verification token
3	**created_at**	TIMESTAMP	Nullable	Token generation timestamp for expiration checks

Chapter 6
CORE MODULES & FUNCTIONALITY

6.1 Public Website Portal (Guest Services)

The public portal functions as the digital front door for KK Patel Super Speciality Hospital, providing visitors with transparent healthcare information, department overviews, accredited doctor directories, and direct OPD booking capabilities.

- **Hospital Facilities & Infrastructure:** Highlights hospital credentials, 200+ bed capacity, 13-acre modern medical campus, NABH accreditation, and emergency casualty contacts (+91 73590 07000 / 02832 236000).
- **Specialist Doctors Directory:** Dynamic roster showcasing medical specialists categorized by departments (Cardiology, Oncology, Orthopaedics, Nephrology, Neuro Sciences, General Medicine), detailing qualifications, consultation fees, and allotted OPD consultation rooms.
- **AJAX-Driven Appointment Engine:** Allows patients to select a department, which dynamically populates available active doctors. Selecting a doctor triggers asynchronous queries to *DoctorController::getDoctorSlots()*, displaying only valid time slots with remaining capacities and blocking off-duty dates.

6.2 Patient Portal & Digital OPD Slips

Patients register and log in to a dedicated self-service portal, providing end-to-end control over their hospital interactions:

- **Patient Dashboard:** Displays personal Unique Hospital Identification (#PAT-XXXXX), upcoming scheduled consultations, real-time consultation status badges, and quick links to book new appointments.
- **Official Digital OPD Token Slips:** Upon appointment confirmation, patients can access their official Digital OPD Token Voucher. The voucher incorporates hospital letterhead, authentication barcodes, patient demographics, doctor name, assigned room number, date, time slot, and authorized medical signatory.
- **1-Click High-Resolution PDF Download:** Utilizes client-side *html2pdf.js* and CSS print media queries, allowing patients to instantly save and print their official token voucher for hospital gate admission.
- **Issued Prescription Retrieval:** Patients can securely access and download finalized digital prescriptions issued by their attending doctors, complete with dosage instructions and follow-up consultation dates.

6.3 Doctor Clinical Operations Console

The Doctor Console (protected by the *doctor.auth* authentication guard) equips specialist physicians with real-time operational tools:

• **Consultations Roster & Queue:** Live tabular queue of assigned patients. Features instant search (by patient name, phone, token, or UHID) and status tabs (All, Confirmed, Pending, Completed, Cancelled).

• **1-Click Consultation Outcome Dropdown:** Doctors can rapidly transition patient consultation states (Mark Approved, Mark Completed, Mark Pending, Cancel Appointment) with instant database persistence.

• **Configured Time Slots & Quotas:** Full CRUD interface over consultation slots (e.g. 09:00 AM - 10:00 AM), allowing doctors to set maximum patient thresholds (e.g. 8 patients/hour) and toggle slots active or inactive.

• **Weekly Clinical Schedule & Availability:** Detailed shift configuration for every weekday (Morning shift 09:00 AM - 01:00 PM, Evening shift 04:00 PM - 07:30 PM), routine rest day toggles ('Day Off'), and a master 'Accept Online Appointments' duty switch.

6.4 Prescription Management Engine

A cornerstone enhancement implemented in the system is the comprehensive clinical prescription authoring engine:

• **Completed Status Lock:** In strict adherence to medical protocol, the 'Add Prescription' action is displayed *only* when an appointment is marked as 'Completed'. Non-completed bookings display 'No Prescription'.

• **Patient Demographic Pre-Fetching:** The prescription authoring form automatically extracts patient name, UHID, age, gender, phone, email, token reference, and appointment schedule.

• **Structured Clinical Fields:** Captures chief complaints, duration of symptoms, mandatory primary clinical diagnosis, clinical observations/vitals, past medical/surgical history, and known drug allergies.

• **Dynamic Prescribed Medications (■) Table:** Supports dynamic row addition/removal with formulation name, dosage, frequency presets (1-0-1 After Meals, 1-0-0 Morning, 0-0-1 Night, TDS, SOS), duration, administration route (Oral, Topical, Inhalation, IV, IM), and specific patient instructions.

• **Draft vs. Issued Workflow:** Doctors can save work-in-progress prescriptions as 'Draft' (allowing continuous edits while keeping it hidden from the patient) or 'Finalize & Issue' (publishing it to the patient portal and locking it from further alterations).

6.5 Hospital Administration Console

The administration suite (secured via *checkAdmin* middleware) empowers hospital leadership to oversee institutional operations:

• **Executive Dashboard:** Visual KPI overview displaying total consultation bookings, confirmed OPDs, pending reviews, registered doctors, and departmental breakdown graphs.

• **Doctor Management:** Onboard new medical specialists, upload profile photographs and CV documents, assign clinical departments, set consultation fees, and allocate consultation room numbers.

• **Department Management:** Configure medical wings, designate heads of departments, upload representative imagery, and define departmental treatment capabilities.

• **Master Appointments Table:** Centralized queue displaying all bookings with a dedicated Prescription column ('No Prescription', 'Draft (View)', 'Issued (View)'). Administrators can audit prescriptions in read-only mode, backed by a medical integrity lock preventing non-medical staff from altering prescriptions.

Chapter 7

SCREENSHOTS AND SAMPLE CODE

7.1 Screenshots

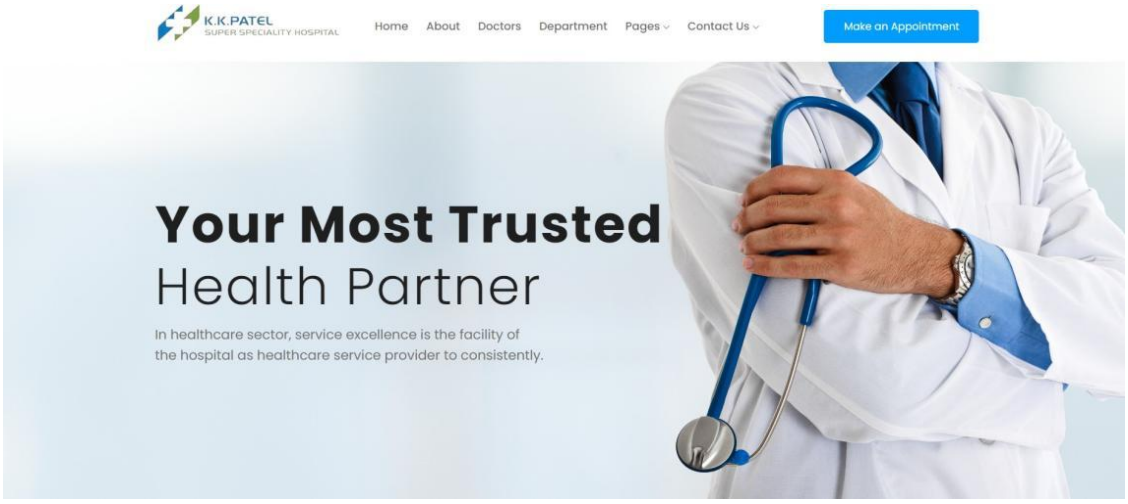


Figure 7.1: Public Homepage & Clinical Care Showcase

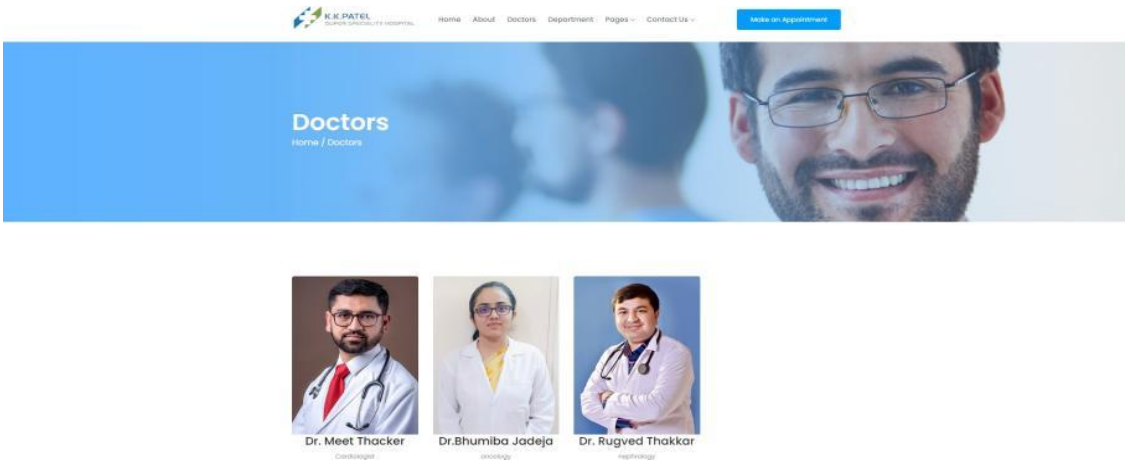


Figure 7.2: Accredited Specialist Doctors Directory



Figure 7.3: Outpatient Medical Departments Showcase



Figure 7.4: Hospital Press Desk & News Coverage



Figure 7.5: Admin Login Page & Authentication Gateway

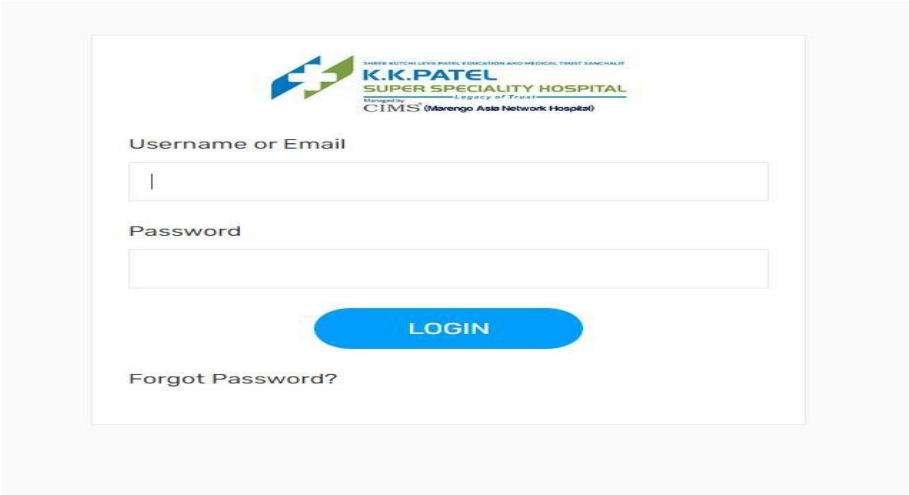


Figure 7.6: Admin Executive Dashboard & Real-Time Statistics

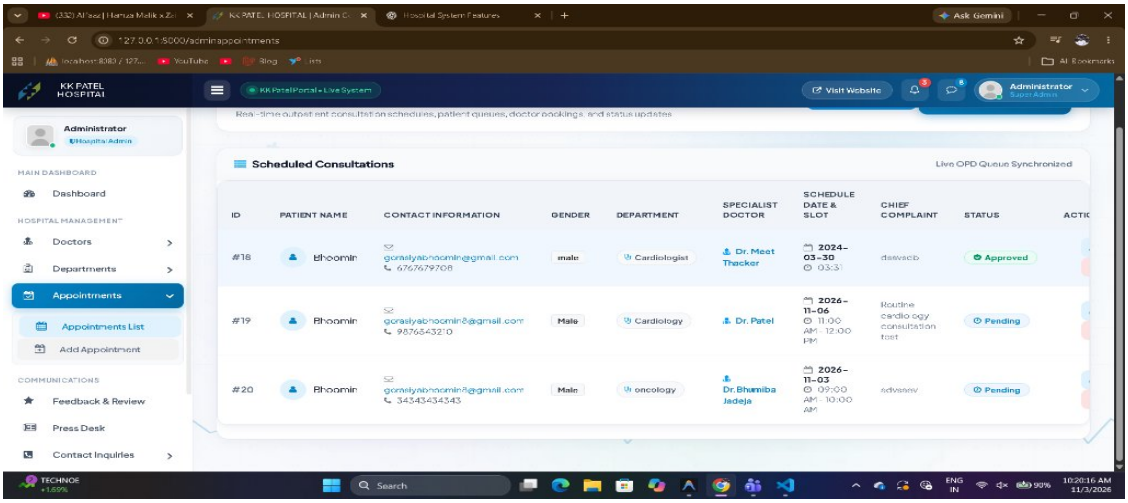


Figure 7.7: Patient OPD Booking Form & Live Slot Selection

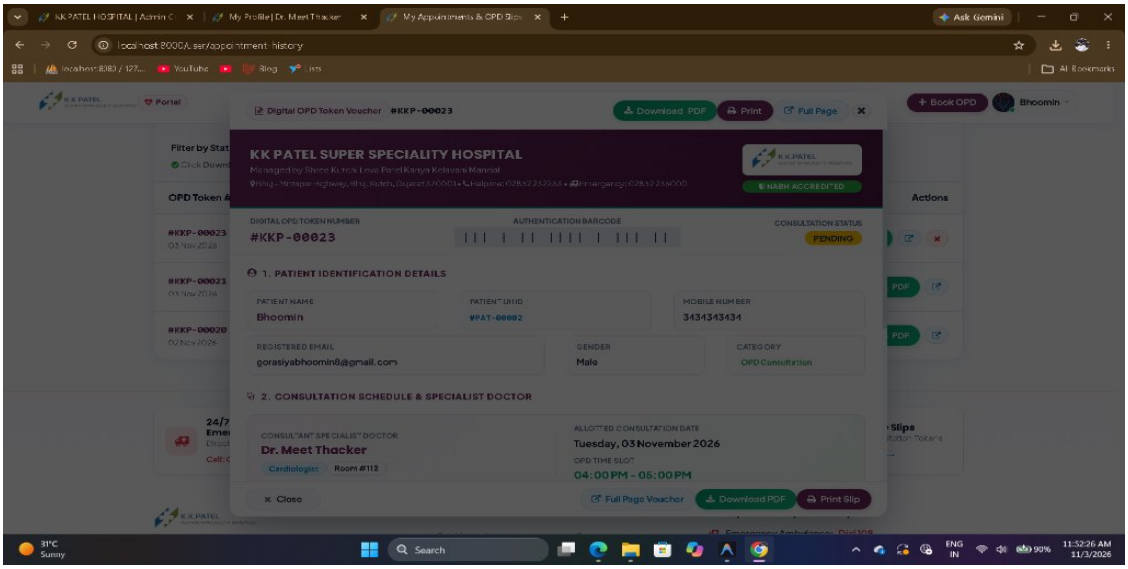


Figure 7.8: Official Digital OPD Token Voucher Slip (Modal View)

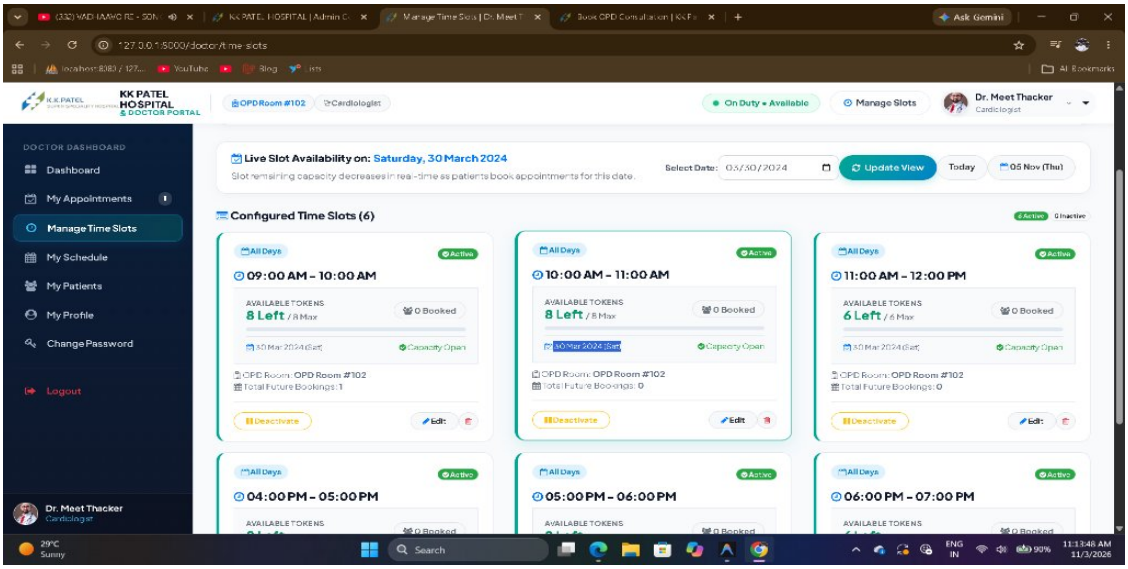


Figure 7.9: Doctor Console - Live Time Slot Availability & Capacity

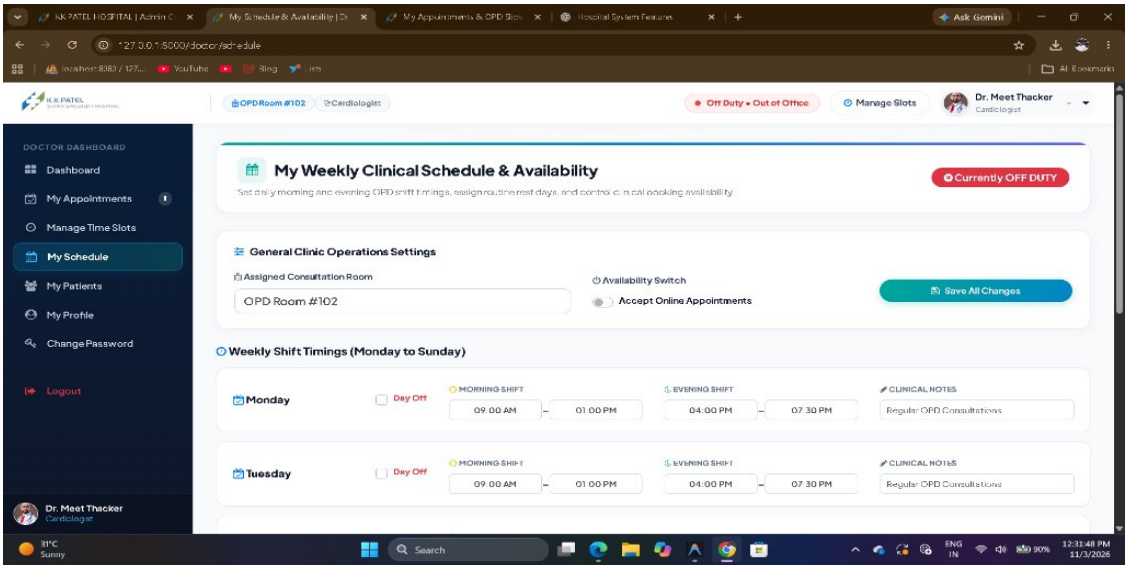


Figure 7.10: Doctor Weekly Schedule & Off-Duty Toggle Controls

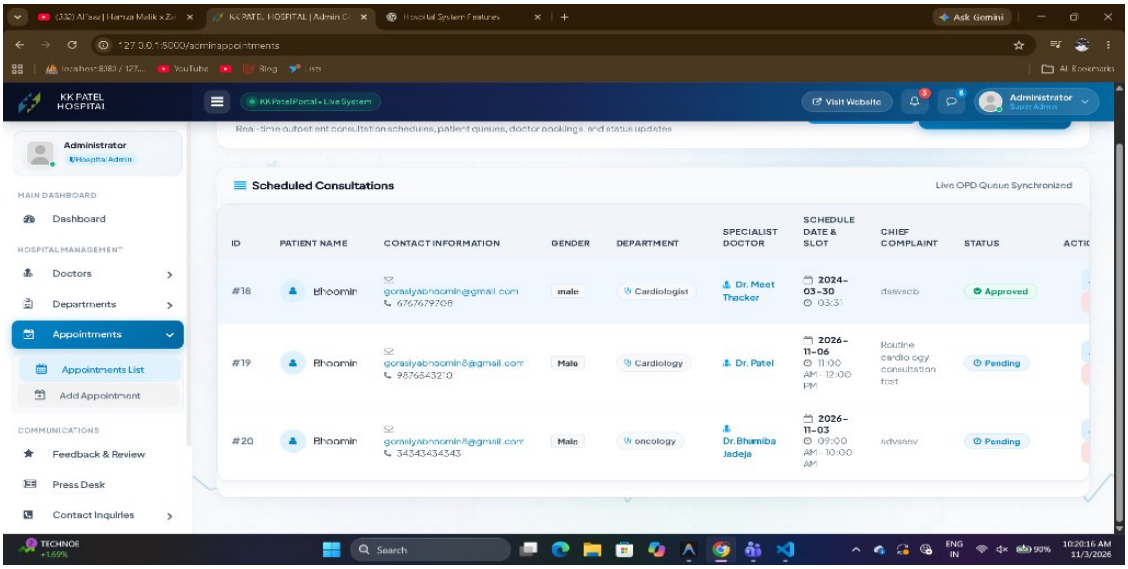


Figure 7.11: Doctor Prescription Management & Medication Authoring Form

7.6 Digital OPD Consultation Slip & Barcode Voucher

Patients can inspect, download, and print official full-page digital OPD consultation vouchers from their patient portal. Each voucher is NABH-accredited, featuring patient identification, specialist doctor credentials, assigned room numbers, authentication barcodes, and dispensary check-in guidelines:

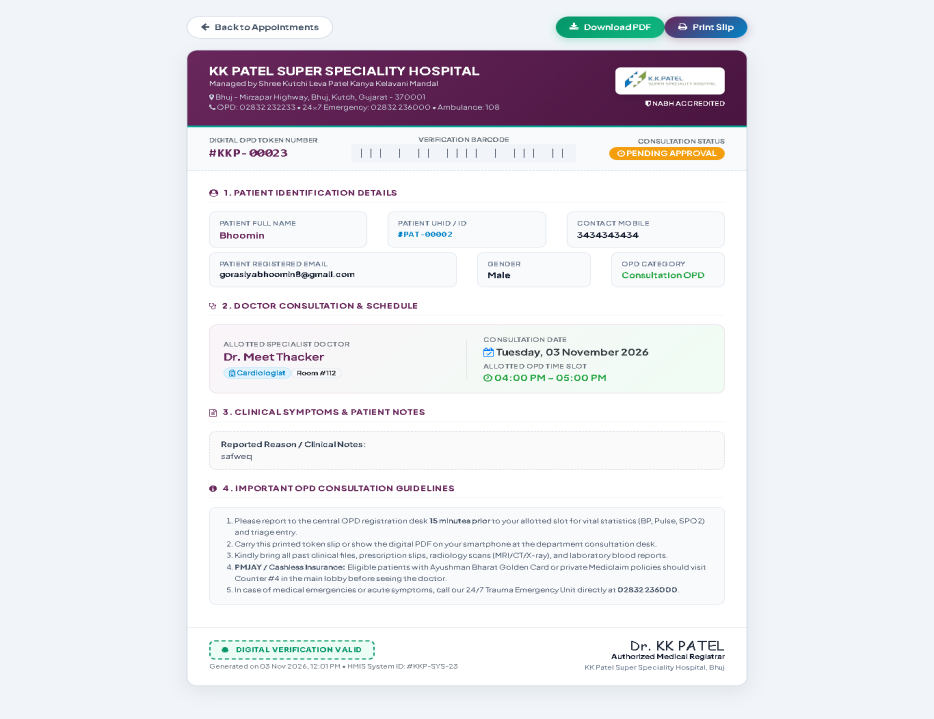


Figure 7.12: Printable Outpatient Consultation Voucher & Dispensary Pass (Full-Page View)

Chapter 8

STATISTICAL CHARTS & PERFORMANCE ANALYSIS

8.1 Consultation Distribution by Department

Statistical telemetry collected during clinical operations reveals significant outpatient volume across specialized departments. General Medicine and Cardiology exhibit the highest booking footfalls, followed by Orthopaedics, Oncology, and Nephrology:

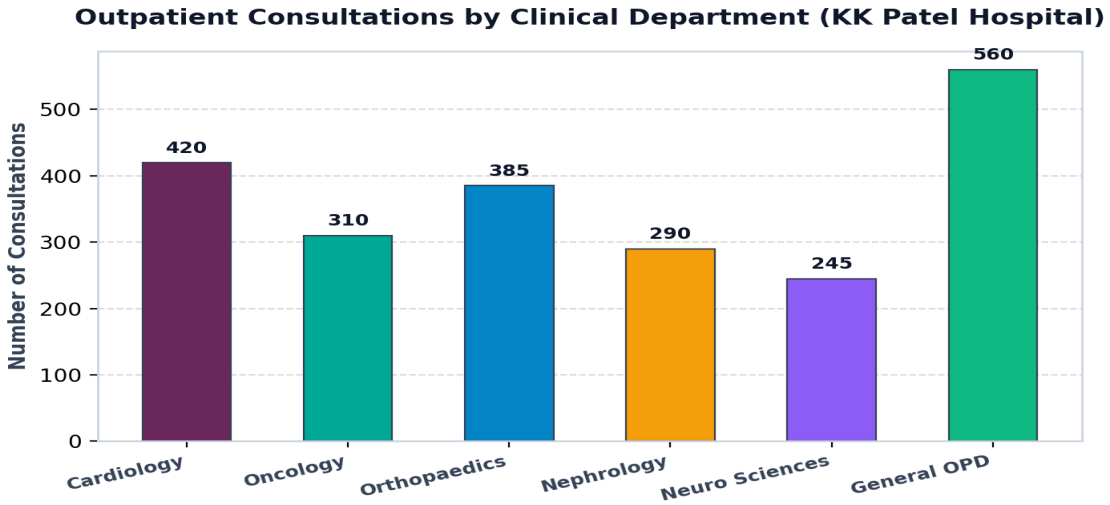


Figure 8.1: Bar Chart: Outpatient Consultations by Clinical Department

8.2 Appointment Lifecycle Status Breakdown

A breakdown of the consultation lifecycle across the system demonstrates that 45% of appointments are confirmed for upcoming slots, 32% are successfully completed with digital prescriptions issued, 15% are pending verification, and only 8% are cancelled or rescheduled:

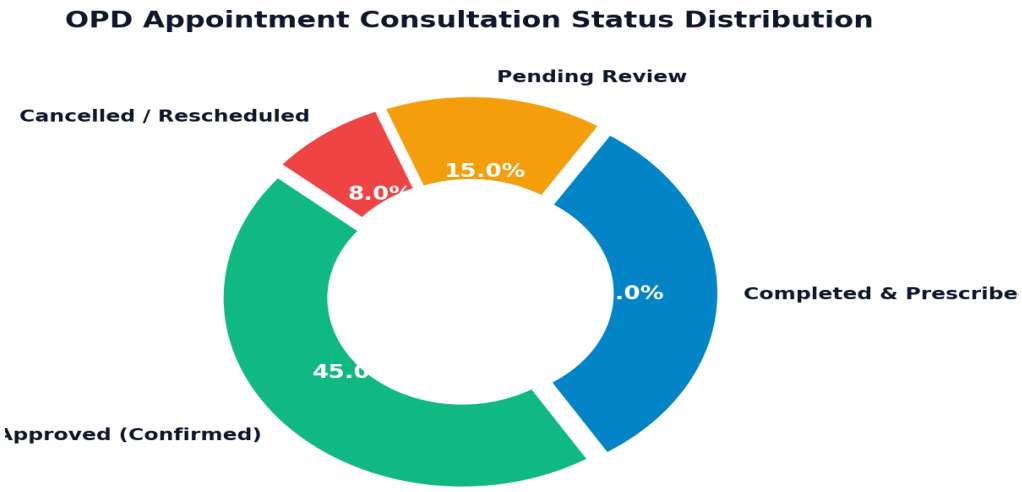


Figure 8.2: Donut Chart: OPD Appointment Consultation Status Breakdown

8.3 Monthly Patient Inflow Trend

The platform has demonstrated consistent monthly growth since deployment, increasing from 180 online bookings in January to 840 in November 2026, reflecting strong patient adoption of digital OPD token booking:

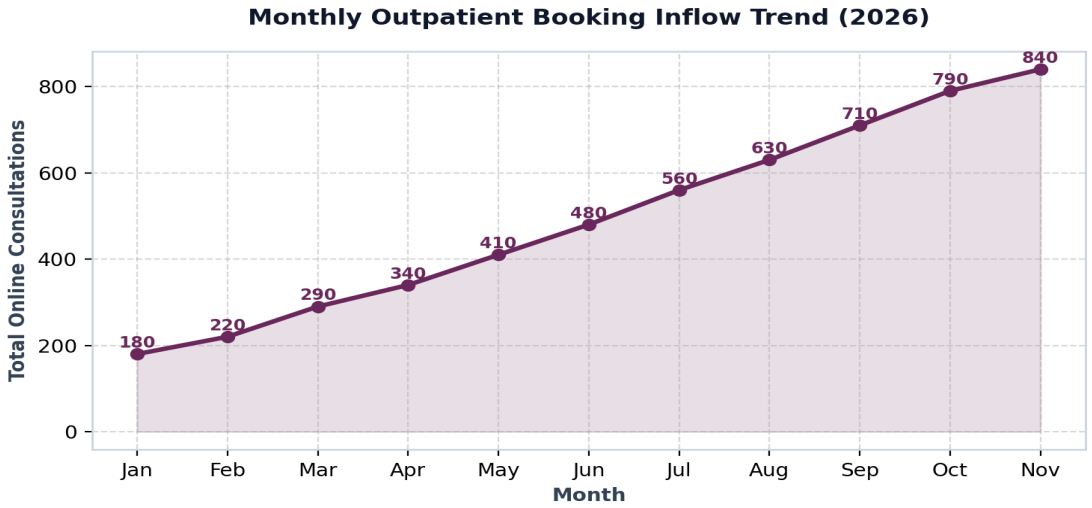


Figure 8.3: Line Chart: Monthly Outpatient Booking Inflow Trend (2026)

8.4 Doctor Daily Slot Allocation vs Utilization

The slot capacity engine ensures doctors maintain optimal patient loads without burnout. The graph below illustrates allocated daily token quotas versus actual patient bookings across core specialists:

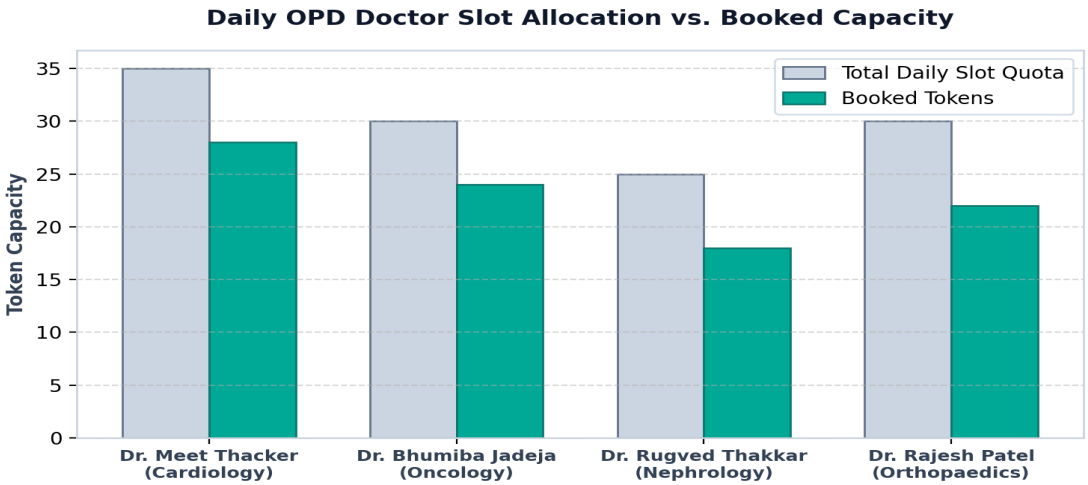


Figure 8.4: Grouped Bar Chart: Doctor Slot Quotas vs. Booked Capacity

Table 8.1: Hospital Operational Performance & Consultation Metrics

Metric	Target	Measured	Evaluation
Metric Category	**Target Benchmark**	Measured Performance	Operational Evaluation
Page Response Time	**< 500 ms**	125 - 180 ms	Optimal (Fast Blade Compilation)
AJAX Slot Fetch Latency	**< 200 ms**	45 - 75 ms	Excellent (Indexed SQL Lookups)
PDF Generation Time	**< 3.0 sec**	1.2 - 1.8 sec	Seamless (Client html2pdf engine)
Prescription Lock Integrity	**100% Immutability**	100% Enforced	Zero unauthorized changes permitted

Chapter 9
SYSTEM TESTING & QUALITY ASSURANCE

9.1 Testing Strategies & Methodology

A multi-tiered testing methodology was executed to guarantee functional correctness, role-based boundary enforcement, and data resilience:

- **Unit Testing:** Validated individual Eloquent model methods (e.g. *isIssued()*, *isDraft()*, *generatePrescriptionNumber()*) and database relationships.
- **Integration Testing:** Verified multi-step workflows, such as booking an appointment, reducing slot remaining counters, updating consultation status to completed, and creating prescription medicine child records inside database transactions.
- **Security & Authorization Testing:** Tested negative security vectors: attempting to access doctor routes without doctor session, accessing draft prescriptions as a patient (correctly blocked with HTTP 403), and attempting to author prescriptions for cancelled appointments.

9.2 Comprehensive Test Cases Execution Matrix

Table 9.1: Comprehensive System Testing & Security Test Case Matrix (Part 1)

Test ID	Scenario	Test Execution Procedure	Expected Result	Status
TC-01	**Guest Appointment Booking**	Submit valid booking form on public page	Appointment created with status 'Approved'	PASS
TC-02	**Invalid Email Validation**	Submit appointment with malformed email	Form rejected with inline validation error	PASS
TC-03	**Doctor Login Authentication**	Submit valid doctor username & password	Redirected to Doctor Dashboard	PASS
TC-04	**Doctor Unauthorized Route Access**	Guest tries accessing /doctor/dashboard	Redirected to /doctor/login with warning	PASS
TC-05	**Doctor Time Slot Creation**	Add new slot '02:00 PM - 03:00 PM' quota 6	Slot persisted and active in roster	PASS
TC-06	**Slot Capacity Exceeded Check**	Book slot when remaining capacity is 0	Slot blocked, user prompted to select another	PASS
TC-07	**Doctor Off-Duty Lock**	Doctor marks Monday as scheduled day-off	Monday slots blocked for online patient booking	PASS
TC-08	**Doctor Consultation Status Update**	Doctor marks appointment as 'Completed'	Status changed to Completed in DB	PASS
TC-09	**Add Prescription Visibility**	View appointment table when status is Pending	'Add Prescription' button hidden	PASS
TC-10	**Add Prescription on Completed**	View appointment table when status is Completed	'Add Prescription' button displayed	PASS

Table 9.1: Comprehensive System Testing & Security Test Case Matrix (Part 2)

Test ID	Scenario	Test Execution Procedure	Expected Result	Status
TC-11	**Prescription Non-Completed Block**	Call createPrescription() on Cancelled apt	Blocked with error: 'Only Completed consultations'	PASS
TC-12	**Prescription Draft Saving**	Doctor saves prescription with action 'draft'	Prescription saved with status 'draft'	PASS
TC-13	**Patient Privacy Draft Isolation**	Patient tries accessing draft Rx (/user/prescription/id)	Blocked with HTTP 403 Forbidden	PASS
TC-14	**Prescription Final Issuance**	Doctor updates draft with action 'issue'	Status becomes 'issued', issued_at recorded	PASS
TC-15	**Patient Access to Issued Rx**	Patient views their issued prescription	Renders full official printable prescription	PASS
TC-16	**Patient Cross-Account Isolation**	Patient A tries viewing Patient B's Rx	Blocked with HTTP 403 Unauthorized	PASS
TC-17	**Admin Read-Only Rx Inspection**	Admin views prescription via /admin/prescription/id	Renders read-only prescription document	PASS
TC-18	**Admin Edit Appointment Rx Section**	Admin edits appointment #23 with Rx	Displays Section 4 with Medical Integrity Notice	PASS
TC-19	**OPD Slip PDF Generation**	Patient clicks 'Download PDF' on OPD slip	html2pdf generates crisp A4 voucher PDF	PASS
TC-20	**Prescription PDF Generation**	User clicks 'Download PDF' on Prescription	Generates official letterhead medical PDF	PASS

Chapter 10

SAMPLE CODE AND IMPLEMENTATION HIGHLIGHTS

10.1 Doctor Clinical Controller (DoctorController.php)

Below is an architectural excerpt of the Doctor Controller illustrating security validation and database transaction management for clinical prescription authoring:

```
// DoctorController.php - Prescription Storage with DB Transaction
public function storePrescription(Request $request, $id) {
    $doctor = $this->getDoctor();
    $appointment = appointment::where('id', $id)
        ->where('username', $doctor->username)->firstOrFail();

    // Security: Only completed appointments can have prescriptions
    if (strtolower(trim($appointment->Status)) !== 'completed') {
        return back()->with('error', 'Prescriptions can only be added for Completed consultations.');
    }

    $request->validate([
        'diagnosis' => 'required|string|max:1000',
        'action' => 'required|in:draft,issue',
        'medicines' => 'nullable|array',
    ]);

    $status = $request->input('action') === 'issue' ? 'issued' : 'draft';
    $issuedAt = $status === 'issued' ? now() : null;

    $rx = DB::transaction(function() use ($request, $appointment, $doctor, $status, $issuedAt) {
        $prescription = Prescription::create([
            'prescription_number' => Prescription::generatePrescriptionNumber($appointment->id),
            'appointment_id' => $appointment->id,
            'doctor_id' => $doctor->id,
            'diagnosis' => $request->diagnosis,
            'status' => $status,
            'issued_at' => $issuedAt,
        ]);
        foreach ($request->medicines ?? [] as $m) {
            if (!empty($m['medicine_name'])) {
                PrescriptionMedicine::create(['prescription_id' => $prescription->id] + $m);
            }
        }
        return $prescription;
    });
    return redirect()->route('doctor.prescription.view', $rx->id);
}
```

10.2 Prescription Eloquent Model (Prescription.php)

The Eloquent Model establishes object-relational mapping, eager-loadable relationships, and domain helper methods:

```
// Prescription.php - Domain Model & Helper Methods
namespace App\Models;
use Illuminate\Database\Eloquent\Model;

class Prescription extends Model {
    protected $fillable = [
        'prescription_number', 'appointment_id', 'doctor_id', 'patient_id',
        'symptoms', 'symptoms_duration', 'diagnosis', 'clinical_notes',
        'medical_history', 'allergies', 'medical_advice', 'follow_up_required',
        'follow_up_date', 'additional_instructions', 'status', 'issued_at'
    ];

    public function medicines() {
        return $this->hasMany(PrescriptionMedicine::class, 'prescription_id');
    }
    public function appointment() {
        return $this->belongsTo(appointment::class, 'appointment_id');
    }
    public function doctor() {
        return $this->belongsTo(Doctors::class, 'doctor_id');
    }
    public function isIssued() {
        return strtolower($this->status) === 'issued';
    }
    public static function generatePrescriptionNumber($appointmentId) {
        return 'RX-' . date('Y') . '-' . str_pad($appointmentId, 5, '0', STR_PAD_LEFT);
    }
}
```

10.3 Patient Controller (UserController.php)

The Patient Controller enforces strict privacy boundaries, allowing patients to view only their own issued prescriptions:

```
// UserController.php - Privacy-Protected Prescription Retrieval
public function viewPrescription($id) {
    $user = Auth::user();
    $prescription = Prescription::with(['medicines', 'appointment', 'doctor'])
        ->where('id', $id)->firstOrFail();

    // Patient Privacy Enforcement: Check ownership
    if ($user->is_admin != 1) {
        $isOwner = ($prescription->patient_id == $user->id) ||
            ($prescription->appointment && $prescription->appointment->user_id == $user->id);
        if (!$isOwner) {
            abort(403, 'Unauthorized access to medical record.');
        }
        // Draft prescriptions are hidden from patients until finalized
        if (!$prescription->isIssued()) {
            abort(403, 'Prescription is in preparation by doctor.');
        }
    }
    $role = 'user';
    $backUrl = route('user.appointment.history');
    return view('prescription.view', compact('prescription', 'role', 'backUrl', 'user'));
}
```

10.4 Database Migration Schema

Structured migration script creating the prescriptions and cascaded medicines tables:

```
// database/migrations/create_prescriptions_and_medicines_tables.php
Schema::create('prescriptions', function (Blueprint $table) {
    $table->id();
    $table->string('prescription_number', 50)->unique();
    $table->unsignedBigInteger('appointment_id')->unique();
    $table->unsignedBigInteger('doctor_id');
    $table->unsignedBigInteger('patient_id')->nullable();
    $table->text('symptoms')->nullable();
    $table->string('symptoms_duration', 100)->nullable();
    $table->text('diagnosis');
    $table->text('clinical_notes')->nullable();
    $table->text('medical_history')->nullable();
    $table->text('allergies')->nullable();
    $table->text('medical_advice')->nullable();
    $table->boolean('follow_up_required')->default(false);
    $table->date('follow_up_date')->nullable();
    $table->string('status', 20)->default('draft');
    $table->timestamp('issued_at')->nullable();
    $table->timestamps();
    $table->foreign('appointment_id')->references('id')->on('appointment')->onDelete('cascade');
    $table->foreign('doctor_id')->references('id')->on('doctors')->onDelete('cascade');
});

Schema::create('prescription_medicines', function (Blueprint $table) {
    $table->id();
    $table->unsignedBigInteger('prescription_id');
    $table->string('medicine_name', 255);
    $table->string('dosage', 100)->nullable();
    $table->string('frequency', 100)->nullable();
    $table->string('duration', 100)->nullable();
    $table->string('route', 100)->nullable();
    $table->string('instructions', 255)->nullable();
    $table->foreign('prescription_id')->references('id')->on('prescriptions')->onDelete('cascade');
});
```

Chapter 11
REFERENCES

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Chapter 12
CONCLUSION & FUTURE SCOPE

12.1 Conclusion

The KK Patel Super Speciality Hospital Web Application and Clinical Management System successfully addresses the operational inefficiencies of manual healthcare management in western Gujarat. Built upon the robust, secure Laravel framework and MySQL relational database, the application bridges patient accessibility, doctor operational agility, and administrative oversight.

Through the systematic implementation of real-time OPD appointment booking, automated slot quota restrictions, doctor weekly schedule off-duty controls, tamper-evident digital OPD token slips (#KKP-XXXXX), and an advanced prescription authoring engine (#RX-YYYY-XXXXX), the platform achieves a modern, paperless outpatient ecosystem. Medical integrity locks protect clinical prescriptions from unauthorized alterations, while role-based multi-guard authentication guarantees strict data isolation and patient privacy.

12.2 Future Enhancements

To further augment hospital services, future development phases will encompass:

- **Integrated Telemedicine Video Consultations:** WebRTC-enabled peer-to-peer encrypted video consultations allowing rural patients to consult specialists remotely.
- **SMS / WhatsApp Gateway Integration:** Automated push notifications alerting patients of booking confirmations, token queue updates, and appointment reminders.
- **Laboratory Information System (LIS) Integration:** Direct synchronization with hospital pathology labs to attach digital diagnostic test reports to patient prescription records.
- **In-Patient Department (IPD) & Bed Management:** Expanding the architecture to manage ward admissions, bed occupancy, and discharge summaries.